SIMULTANEOUS EQUATIONS: FINDING THE SECRET CROSSING

The Secret Crossing Point Where Two Lines Meet – with Rahul & Amit

The Treasure Map: Our Grid as a Neighborhood!

Imagine the grid as a map — just like the neighborhood you know! 🗺️
Every point on the grid is like a house or a landmark.
To reach any point, we move only forward (right) and upward — using whole numbers only!

📍 Start at zero — that's our home base.
➡️ Move right to count along the bottom.
⬆️ Move up to count along the side.

"Every crossing point on the grid is a special address — and today, we'll find the most secret one of all!"

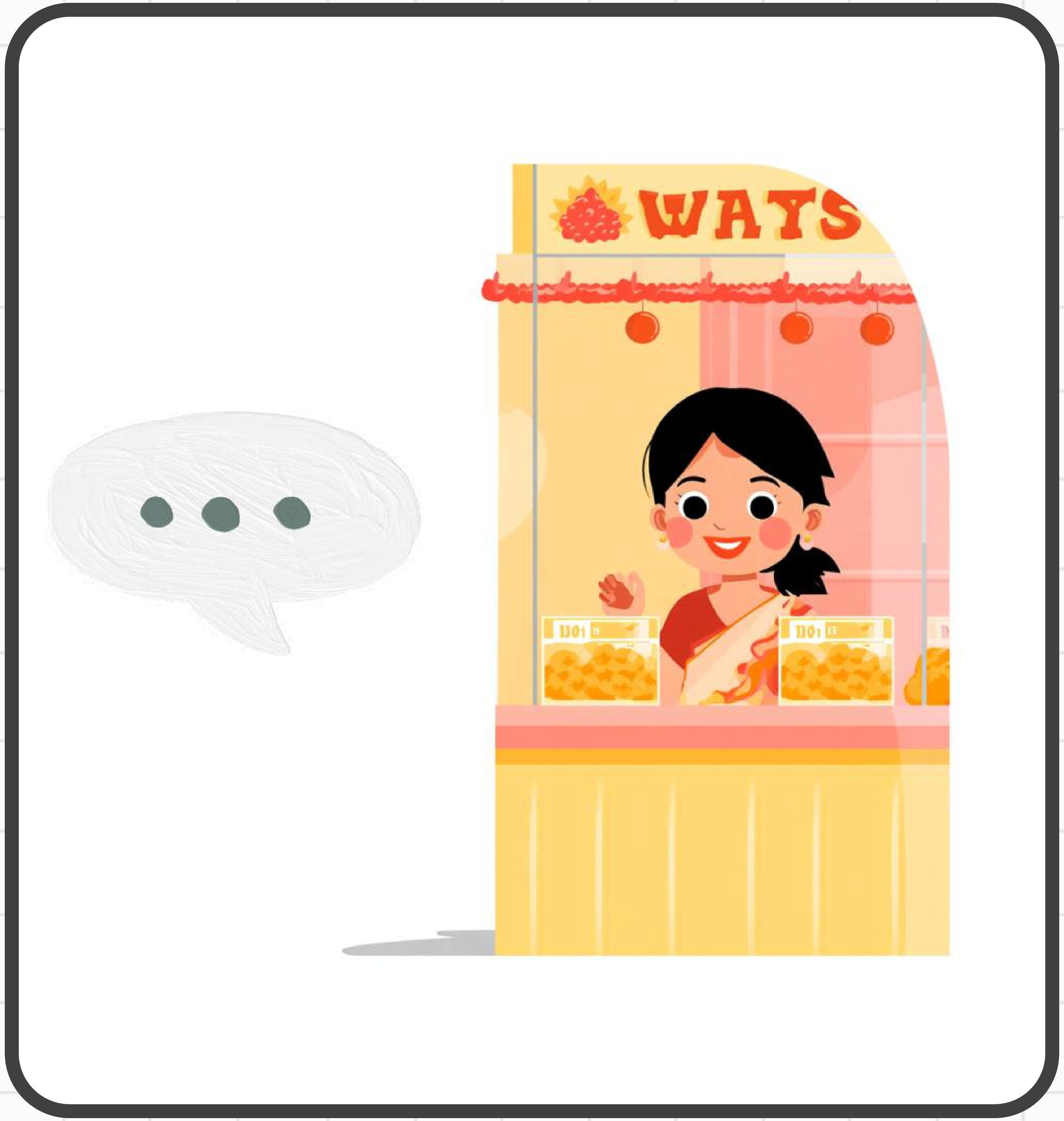
Lines are Rules: The Sweet Shop Puzzle

Priya has ₹10 to spend on Jalebis and Laddoos.
Each line on the grid shows ALL the ways she can spend exactly ₹10!

Here are some ways Priya can spend ₹10:

- 0 Jalebis + 2 Laddoos = ₹10
- 1 Jalebi + 1 Laddoo = ₹10
- 2 Jalebis + 0 Laddoos = ₹10

Every combination is a point on the grid.
Connect the points and you get a RULE LINE! 🌟

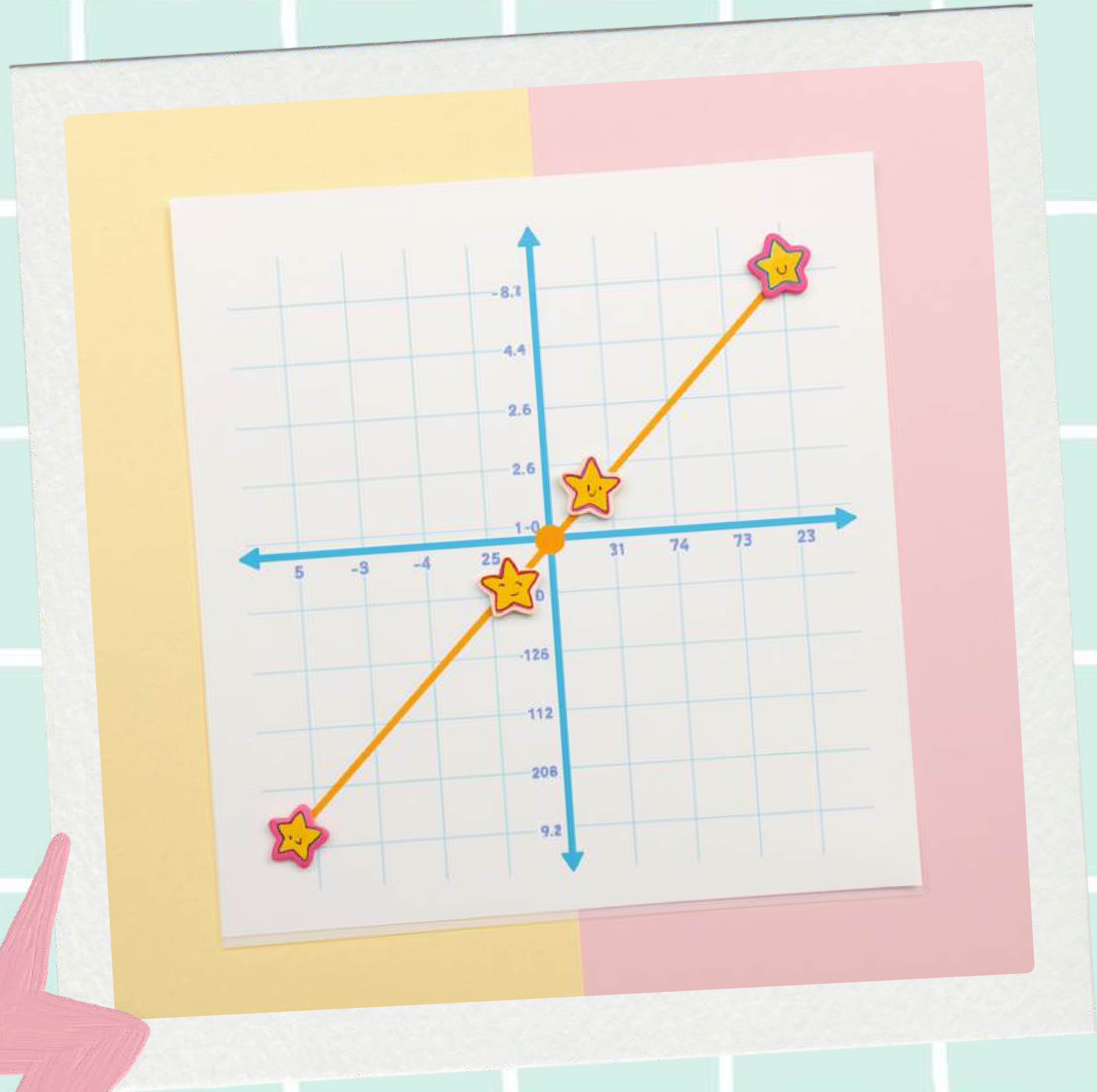


Plotting the Sweet Shop Rules! 🍬

Each point on the grid tells a story of how Priya can spend ₹10 on Jalebis and Laddoos!

📍 (0, 5) → 0 Jalebis + 5 Laddoos = ₹10
📍 (2, 4) → 2 Jalebis + 4 Laddoos = ₹10
📍 (4, 3) → 4 Jalebis + 3 Laddoos = ₹10
📍 (6, 2) → 6 Jalebis + 2 Laddoos = ₹10
📍 (8, 1) → 8 Jalebis + 1 Laddoo = ₹10

✨ Connect the dots — they form the Rule Line! This line shows ALL the right ways to spend ₹10 on sweets!



RULE LINE

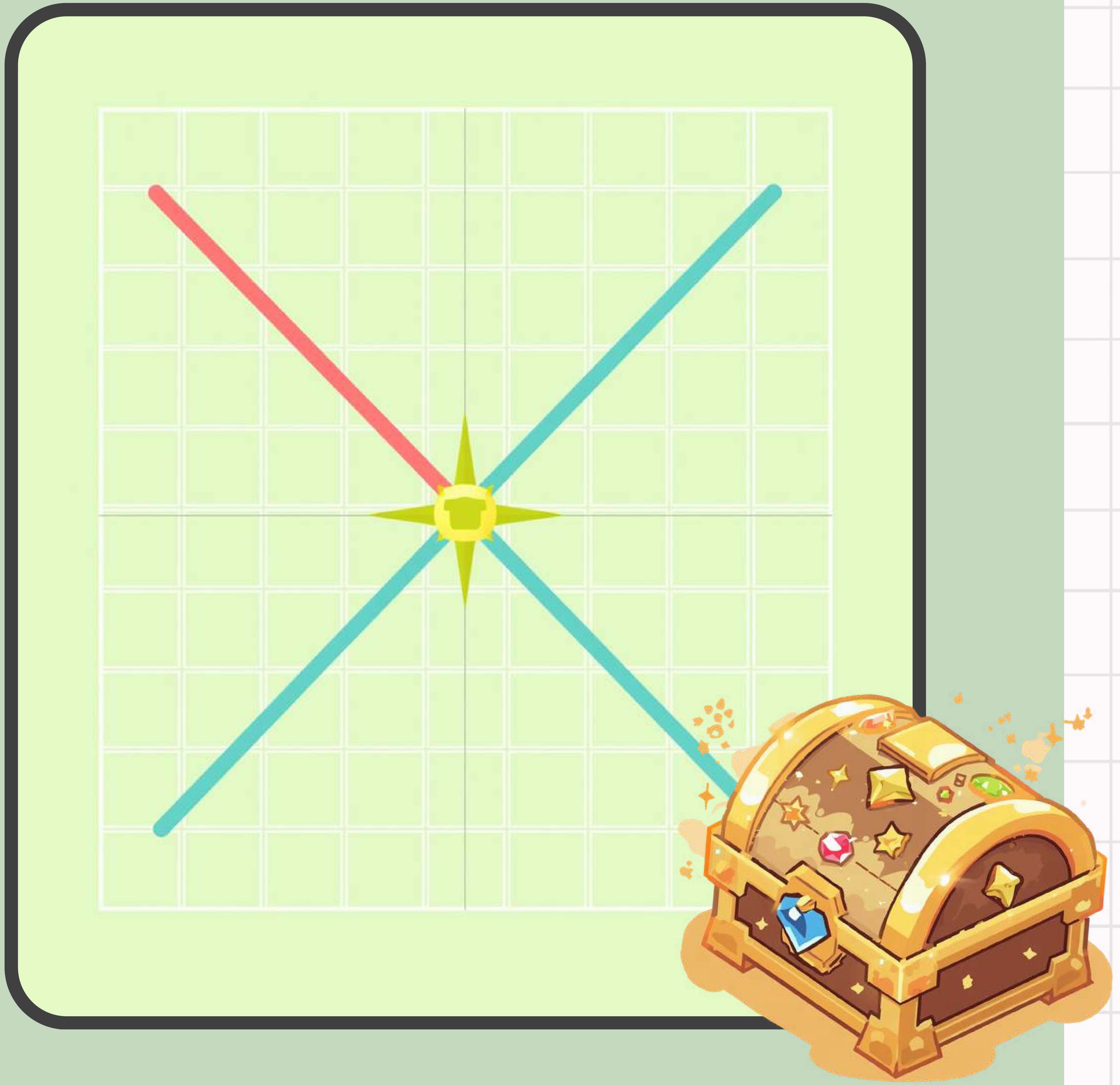
The Sweet Shop Grid

Another Rule Joins the Game!

Rahul also has ₹10 to spend at the market!
Here are all the ways Rahul can buy Samosas and Kites:

- 0 Samosas + 2 Kites = ₹10
- 1 Samosa + 1 Kite = ₹10
- 2 Samosas + 0 Kites = ₹10

Each combination makes a point on the grid — connect them and you get Rahul's Rule Line!



The Secret Crossing Point!

The crossing point is the special place where BOTH rules are true at the same time!
When two lines meet on the grid, that one magical point is the answer — it follows Rule 1 AND Rule 2 together.
Rahul and Priya both get exactly what they want at this point. It's like finding hidden treasure on a map!

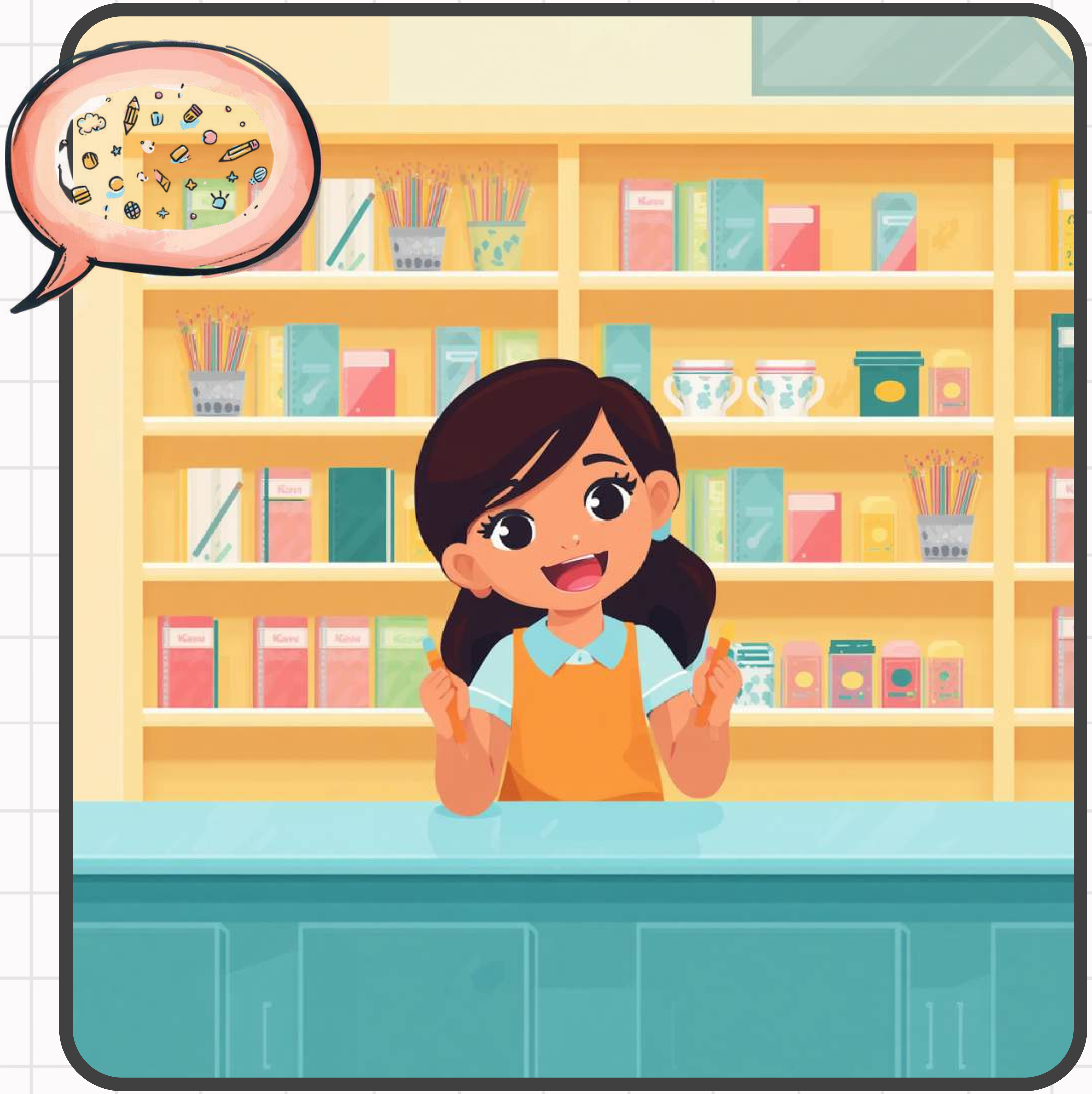
✦ Look for where the two lines cross
✦ Mark it with a star or treasure chest
✦ That point is your secret answer!

Story Time: Stationery Shop Challenge

Kavya has ₹10 to spend at the stationery shop!
Pencils cost ₹2 each and Erasers cost ₹1 each.
Here are some ways Kavya can spend exactly ₹10:

- 0 Pencils + 10 Erasers = ₹10
- 1 Pencil + 8 Erasers = ₹10
- 2 Pencils + 6 Erasers = ₹10
- 3 Pencils + 4 Erasers = ₹10
- 4 Pencils + 2 Erasers = ₹10
- 5 Pencils + 0 Erasers = ₹10

Each combination is a point on our grid.
Connect the dots and you get Kavya's Rule Line!



Kavya's Line on the Grid!

Kavya has ₹10 to spend on pencils and erasers. Let's plot all her choices!
✏️ 0 Pencils + 5 Erasers = ₹10
✏️ 1 Pencil + 4 Erasers = ₹10
✏️ 2 Pencils + 3 Erasers = ₹10
✏️ 3 Pencils + 2 Erasers = ₹10
✏️ 4 Pencils + 1 Eraser = ₹10
✏️ 5 Pencils + 0 Erasers = ₹10
Each point is a way to spend ₹10 on pencils and erasers.
Connect the dots — they form Kavya's Rule Line! 🌟

Kavya's Rule Line

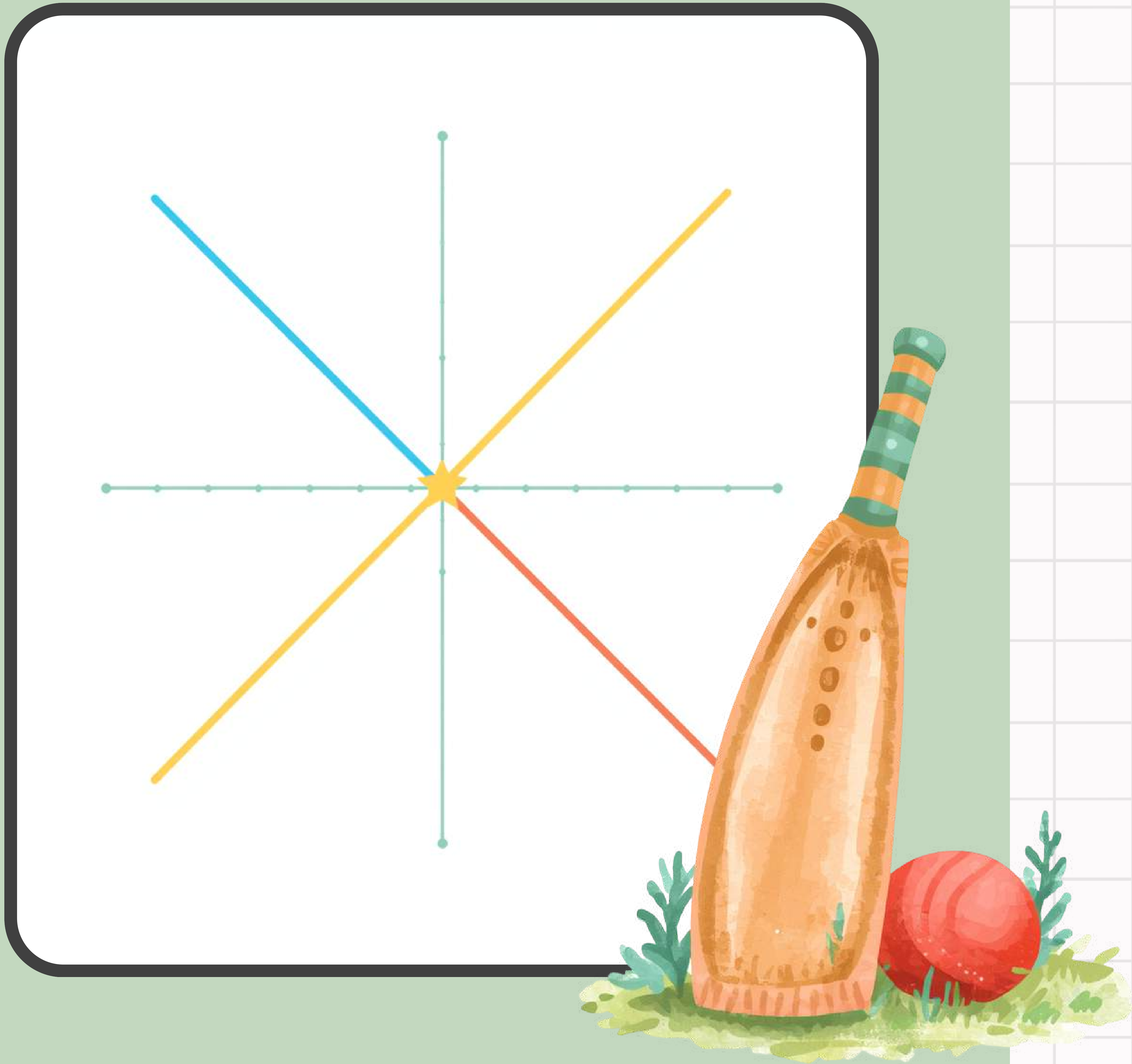
Guided Practice: Cricket Scores!

Rahul and Amit are playing cricket. Together, their runs add up to a total score!

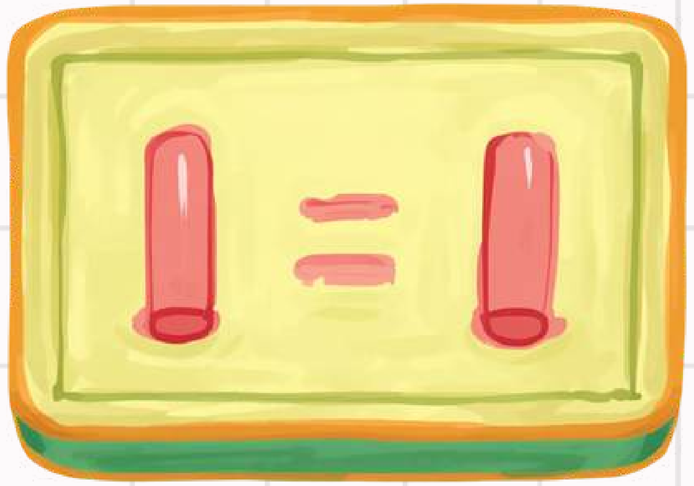
🏏 Each rule is a LINE showing all the possible run combinations:

- Rahul's runs + Amit's runs = 10
- Rahul's runs + Amit's runs = 8

Plot both lines on the grid. Where do they CROSS? That secret crossing point tells us the exact runs each player scored!



Cricket Score Grid!



Watch as Rahul's and Amit's run rules come to life on the grid!

Each colourful line shows all the possible run combinations for one player. Where the two lines cross — that is the Secret Crossing Point!

⭐ The crossing point shows the winning score for both Rahul and Amit together.

"Both rules are true at the same time — right here at this special point!"

Quick Quiz: Find the Crossing!

Question 1 🏏

Rahul scores 4 runs and Amit scores 6 runs. Look at the grid — where is the crossing point? Write the coordinates!

Question 2 📏

Two rules make two lines on the grid. Line 1: $x + y = 10$. Line 2: $x + y = 6$. Do these lines ever cross? Why or why not?

Question 3 ⭐

Priya spends ₹10 on Jalebis and Laddoos. Rahul spends ₹10 on Samosas. Find the secret crossing point where both rules are true!

What Did We Learn Today?

The Grid is a Map

The grid is like a map of a neighborhood. We move forward and right using whole numbers to find any point!

Lines are Rules

Every line on the grid shows all the ways to follow one rule — like spending ₹10 on sweets or stationery!

The Crossing Point

The secret crossing point is where two rule lines meet — the one special place where both rules are true at the same time!

Thank You & Questions!

Thank you for joining our Secret Crossing Adventure today!
You did a fantastic job learning how two lines meet at a special crossing point on the grid.
Remember — every crossing point is a secret waiting to be discovered!